The Physical Layer

Lecture 29

Computer Networks and Distributed Programming

What lies under the frame

- Everything so far ended with a frame handed to the adapter. Today: what the adapter does with it.
- Every other layer adds a header. This one turns bits into a signal — a voltage, a light, a radio wave — and is the only one that moves anything.
- Which medium: copper, glass or air, and what each can carry.
- Which signal: how a bit becomes a signal, and how the receiver turns it back into a bit.
- How much: every channel has a ceiling on its bit rate, and nature, not engineering, decides where it is.
- How many at once: laying a cable is expensive, so one cable must carry many conversations.

Guided and unguided media

- On every hop a bit is transmitted again, by one transmitter and one receiver, across a physical medium.
- The medium changes along the path: copper to the switch, fibre to the provider, radio to the phone in your pocket.

Two families

Guided media: the wave is guided along a solid medium — twisted-pair copper wire, coaxial cable, optical fibre.

Unguided media: the wave propagates through the air or through space — a wireless LAN, a cellular network, a satellite channel.

- The cable is cheap, laying it is not: labour costs orders of magnitude more than material, so buildings are wired for more than they need.

The fastest link is a truck

A box of tapes, 800 terabytes, delivered anywhere in the country in 24 hours. What is its bit rate?

6400 terabits in 86 400 seconds: about 74 Gb/s. An hour away by road, 1.8 Tb/s. No network link in the building comes close.

- It is done: Amazon moves whole datacenters in a truck carrying 100 PB of disks.
- The catch: the first bit arrives a day after it left. The rate is enormous, the delay is measured in hours.
- Rate and delay are independent, as transmission and propagation delay were. A medium is judged on both.
- Everything from here on is about media that answer in microseconds.

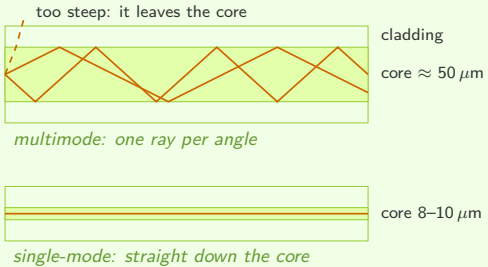
Twisted pair

- Two insulated copper wires, about 1 mm thick, twisted together: the oldest medium, and still the most common.
- Two parallel wires are a fine antenna. Twisted, the waves of successive twists cancel, and the pairs in a bundle stop interfering.
- The signal is the difference between the two wires: noise hits both alike, and the difference survives it.
- Categories: Cat 5e, four pairs in a sheath; Cat 6A, 10 Gb/s to 100 m; Cat 8, faster but only for about 30 m.
- 100 Mb/s Ethernet uses two pairs, one per direction. 1 Gb/s uses all four, in both directions at once.
- The pair that reaches your home carries the telephone, and ADSL on top of it.

Coaxial cable, and even power lines

- A copper core, an insulator, a braided outer conductor, a sheath: two conductors, concentric instead of parallel.
- The shielding buys more bandwidth and far better noise immunity than an unshielded pair: up to about 6 GHz.
- Cable television and cable Internet shift each signal into its own frequency band, and many bands share one cable.
- It can be a shared medium: every station attached hears what the others send.
- Power lines: data superimposed on the 50 Hz power signal. The wiring was never meant for megahertz: it attenuates, radiates, and changes whenever an appliance switches on.

Light trapped in glass



- A light source, a glass fibre, a detector: a pulse of light is a 1, darkness a 0. The system is simplex.
- Light meeting the core's boundary above the critical angle is reflected back in, and travels kilometres with almost no loss.
- Multimode: a wide core, many rays at many angles. Single-mode: a core a few wavelengths wide: straight, 50 times farther — 100 Gb/s for 100 km, unamplified.

Fibre against copper

- Bandwidth: three wavelength bands are used, around 0.85, 1.30 and 1.55 μm, and each is 25 000 to 30 000 GHz wide.
- Attenuation: repeaters every 50 km on long lines, against about 5 km on copper.
- Immunity: no electromagnetic interference, no power surges; light does not leak, so a fibre is hard to tap.
- Weight: a thousand twisted pairs 1 km long weigh 8000 kg; two fibres carry more, and weigh 100 kg.
- Against it: it breaks if bent too far, it is one-way — two fibres per link — and its interfaces cost more.
- About 100 Gb/s per channel is not the limit of the glass: it is how fast we convert between electricity and light.

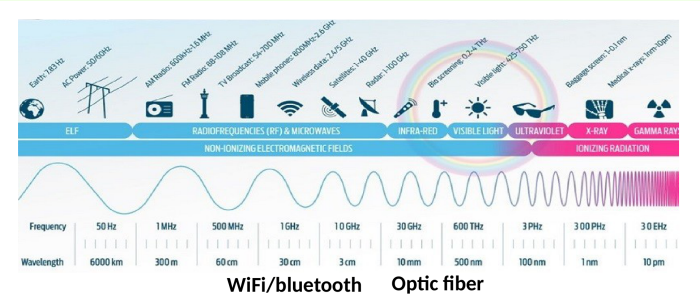
Three media, three trade-offs



Type	Signal	Distance	Bandwidth	Weak point
Wireless	radio	30–50 m	1.3 Gb/s (WiFi 5) to 9.6 (WiFi 6), shared	security, speed
Twisted pair	electricity	100 m; 30 m	10 Gb/s (Cat 6A); 25–40 (Cat 8)	interference
Optical fibre	light	100–200 km per span	1–400+ Gb/s per wavelength	fragility

- The boundaries move every few years: read the figures as orders of magnitude, not as specifications.
- Radio figures are peak rates shared by the whole cell; a laptop with two antennas sees a fraction of them.

The electromagnetic spectrum



- Radio, microwaves, infrared and light are one phenomenon, differing only in frequency: $\lambda f = c$.
- 2.4 GHz is a wave 12.5 cm long; a fibre at 1.55 μm, about 190 THz. WiFi sits between the 1 and 10 GHz ticks, not under its label.
- In copper and glass the speed is about 2/3 of c: the 2×10^8 m/s behind every propagation delay we computed.

What the frequency decides

- Low frequencies: pass through walls and follow the ground, but carry little: AM radio.
- High frequencies: travel in straight lines, bounce off obstacles, and above about 4 GHz are absorbed by rain.
- Path loss: a radio wave spreads over a growing surface, so its power falls at least as $1/r^2$.
- Reflected copies arrive late and out of phase, and can cancel the direct wave: multipath fading.
- A cable loses the same fraction every metre; radio loses it every time the distance doubles. Radio reaches far — and interferes far.
- Hence licences for almost all of the spectrum. Towards infrared, waves behave like light: they stop at the wall.

Unlicensed bands, and why they spread

- A few bands are open to anyone, the ISM bands: 2.4 GHz almost everywhere, 5 and now 6 GHz for WiFi.
- The price of no licence: a cap on transmit power, and the duty to spread the signal over many frequencies.
- Frequency hopping: the transmitter jumps frequency hundreds of times a second: Bluetooth. Hard to jam, and one bad frequency does not stop it.
- Direct sequence: a code spreads every bit over a wide band, and different codes can share it — the idea behind CDMA.
- So your WiFi, your mouse and the microwave oven share 2.4 GHz, and interfere by design.

A signal is a sum of sines

- On a wire, bits travel as a physical quantity that changes with time — a voltage, a light intensity: a function $g(t)$.

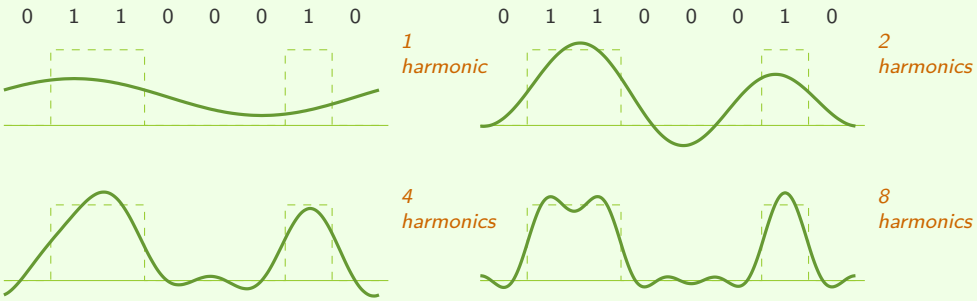
Fourier series

Any reasonably behaved periodic function of period T is a sum of sines and cosines, with $f = 1/T$:

$$g(t) = \frac{c}{2} + \sum_{n=1}^{\infty} a_n \sin(2\pi nft) + \sum_{n=1}^{\infty} b_n \cos(2\pi nft)$$

- f is the fundamental, the n-th term the n-th harmonic. A finite message is treated as if it repeated forever.
- What matters here: a channel treats each harmonic separately, passing some almost untouched and attenuating others.

What a channel does to eight bits



- The letter b, 01100010, and what arrives when the channel passes only the first 1, 2, 4 or 8 harmonics.
- Every medium passes frequencies up to a cutoff and weakens the rest: unequal attenuation is distortion.
- With 8 harmonics the bits can be read. More would only make the square prettier.

Narrower band, fewer bits

- A voice-grade telephone line cuts off just above 3000 Hz.
- Sending 8 bits at R b/s, the first harmonic is at $R/8$ Hz: the line passes about $24\,000/R$ harmonics.
- At 9600 b/s the signal looks like the two-harmonic panel; at 38 400, nothing gets through.
- Limiting the bandwidth limits the bit rate, even on a perfect channel. More levels will help — within limits.

b/s	1st harm.	passed
300	37.5 Hz	80
1200	150 Hz	20
2400	300 Hz	10
4800	600 Hz	5
9600	1200 Hz	2
19 200	2400 Hz	1
38 400	4800 Hz	0

Bandwidth in hertz, bandwidth in bits per second

- At the start of the course bandwidth was a rate. We said then that down here it would also be a width.
- Analog bandwidth: the band of frequencies a medium passes without strong attenuation, in Hz. A property of the wire: its material, thickness, length.
- Quoted up to where the received power has halved; filters narrow it on purpose — an 802.11 channel is 20 MHz or more.
- Digital bandwidth: the maximum bit rate, in b/s: what is obtained from the analog one. The next frames give the exchange rate.
- Baseband: from 0 up to some frequency, as on most wires. Passband: shifted into a higher band, as all radio is.
- What carries information is the width of the band, not where it sits.

Nyquist: the ceiling of a perfect channel

Nyquist, 1924

A noiseless channel of bandwidth B Hz, carrying a signal with V discrete levels, cannot exceed

$$R_{\max} = 2B \log_2 V \text{ b/s}$$

- Behind it: a signal filtered to B Hz is completely rebuilt from $2B$ samples a second. Sampling faster finds nothing new.
- A binary signal on a 3 kHz channel: at most 6000 b/s, however good the electronics.
- The formula rewards levels: 4 double the rate, 16 quadruple it, and nothing in it says when to stop.
- It holds for a perfect channel. No channel is perfect.

Signal, noise and decibels

- Every channel carries thermal noise, from the motion of the molecules in it. What matters is its ratio to the signal, S/N.
- The ratio spans many orders of magnitude, so it is quoted on a logarithmic scale, in decibels: $10 \log_{10}(S/N)$.

S/N	2	10	100	1000	10 000
dB	3	10	20	30	40

- 3 dB is a factor of two: the half-power point at which analog bandwidth is quoted.
- Losses in dB simply add: a twisted pair loses about 20 dB every 100 m, radio about 6 dB every time the distance doubles.

Shannon: the ceiling of a real channel

Shannon, 1948

A channel of bandwidth B Hz with thermal noise and signal-to-noise ratio S/N cannot exceed

$$C = B \log_2(1 + S/N) \text{ b/s}$$

whatever the number of levels, whatever the sampling.

- ADSL: about 1 MHz of line and 40 dB on a short loop: $10^6 \log_2 10\,001 \approx 13$ Mb/s. The original standard promised 12.
- Beyond that there are two ways only: more bandwidth or less noise. ADSL2+ doubled the band.
- A system that claims more belongs with the perpetual motion machines.

Why not simply use more levels?

Nyquist says every doubling of the levels adds a bit per symbol. Why not a thousand levels?

Because of noise. Levels closer together than the noise can no longer be told apart. Setting Nyquist's rate equal to Shannon's capacity,

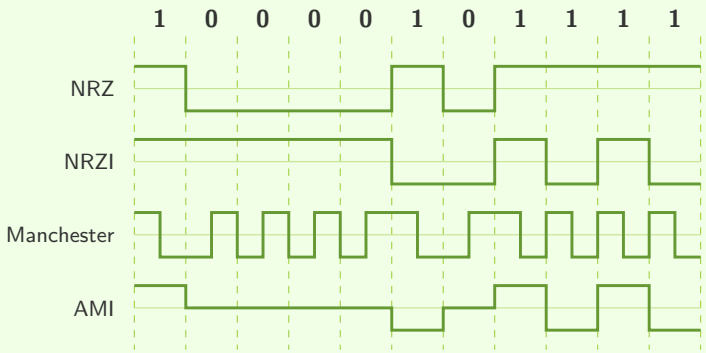
$$2B \log_2 V = B \log_2(1 + S/N) \quad \Longrightarrow \quad V = \sqrt{1 + S/N}$$

- At 40 dB that is about 100 levels; at 20 dB, only 10.
- The two answer different questions: Nyquist how fast to signal, Shannon how many bits each signal can carry.
- Everything that follows — line codes, constellations — is a way of spending that budget.

Line codes: from bits to levels

- Baseband: a bit becomes a level directly: one voltage for 1, another for 0, or light and dark on a fibre. This is the NRZ we met when looking for frame boundaries.
- The receiver samples at regular intervals and takes each sample for the closest level: what arrives is attenuated and noisy.
- NRZ can change at every bit, so at R b/s it needs at least $R/2$ Hz: Nyquist, with two levels.
- Four levels carry 2 bits per symbol: the symbol rate halves, and so does the bandwidth needed.
- Baud: the old name of the symbol rate. Bit rate = symbol rate $\times$ bits per symbol, and the two are often confused.
- NRZ alone is rarely used: line codes reshape it.

Four line codes



- The same eleven bits. NRZ: the level is the bit. NRZI: a 1 is a transition, a 0 is none.
- Manchester: a transition in the middle of every bit, its direction the bit. AMI: a 0 is zero, the 1s alternate.

Keeping the receiver's clock

- With NRZ a run of equal bits is a flat line: fifteen zeros look like sixteen, unless the two clocks agree to a fraction of a bit.
- A separate clock wire would do it — but a second wire is better spent on data.
- Manchester: the clock XORed into the data: a transition in every bit, for the receiver to lock onto. The price is twice the bandwidth. Classic Ethernet.
- NRZI: a transition for one value, none for the other. Runs of the first are safe; runs of the second are not.
- USB uses NRZI with a 0 as the transition, so the danger is a long run of ones — broken by the bit stuffing we saw with framing: a 0 forced in after six 1s.

4B/5B

4B	5B	4B	5B
0000	11110	1000	10010
0001	01001	1001	10011
0010	10100	1010	10110
0011	10101	1011	10111
0100	01010	1100	11010
0101	01011	1101	11011
0110	01110	1110	11100
0111	01111	1111	11101

- Every 4 data bits leave as 5, from a fixed table chosen so that there are never more than three 0s in a row.
- Sent with a code that marks every 1 by a transition, the line is never flat for long.
- The cost is 25%, against Manchester's 100%.
- 16 of the 32 patterns are left over: one means an idle line, another the start of a frame — the coding violations that delimited frames.

Scrambling, and the killer packet

- Scrambling: XOR the data with a pseudorandom sequence; the receiver XORs again with the same sequence and has the data back.
- No overhead at all, and the energy is spread over all frequencies instead of peaking at a few.

What if the data happen to be equal to the scrambler's sequence?

They XOR to all zeros: a flat line, and a lost clock. Early IP over SONET used a short, predictable sequence, and a user could send killer packets that brought the link down.

- Scrambling makes long runs unlikely, not impossible. It is safe only with a long sequence that nobody can predict.

Balanced signals

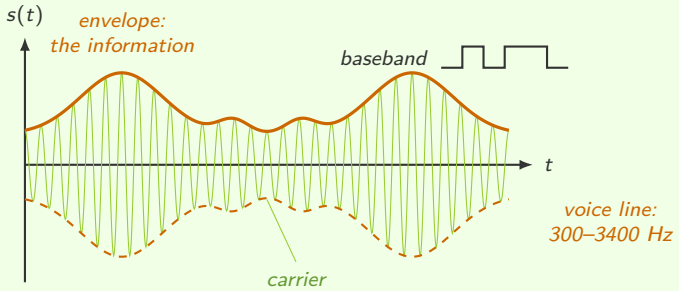
- A signal with as much positive voltage as negative, even over short stretches, has no DC component.
- It matters: coaxial cable and transformer-coupled lines filter DC out, and the energy put into it is wasted.
- The average of a balanced signal is a ready-made decision threshold; an unbalanced one drifts, and symbols are misread.
- AMI: a 0 is zero and successive 1s alternate $+$ and $-$: balance bought with a third level.
- 8B/10B: 8 bits as 10, choosing each symbol or its complement to cancel the running disparity: never more than 2 out of balance, never more than five equal bits.

Every Ethernet speed, its own encoding

Ethernet	Medium	Line code	On the wire
10 Mb/s, classic	coax, twisted pair	Manchester	20 MHz: two per bit
100BASE-TX	2 pairs, Cat 5	4B/5B	125 MHz
1000BASE-X	fibre	8B/10B	1.25 Gbaud
1000BASE-T	4 pairs, Cat 5	5 levels, 2 bits	125 Msymbol/s per pair
10 Gb/s, fibre	fibre	scrambling + 64B/66B	3% overhead

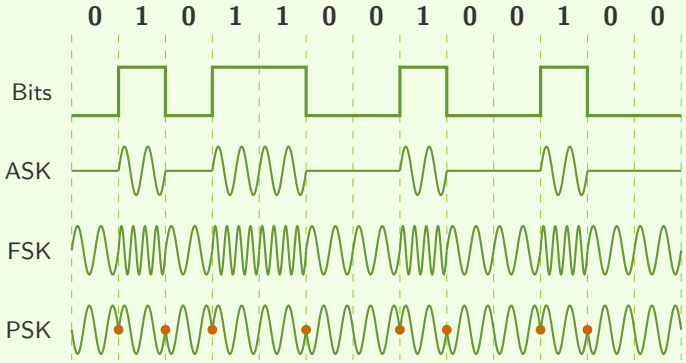
- 1000BASE-T: 4 pairs $\times$ 125 Msymbol/s $\times$ 2 bits $=$ 1 Gb/s, with every pair sending and receiving at the same time.
- The frame format never changed. What changed at every generation is how bits become a signal.

Modulation: a signal riding a carrier



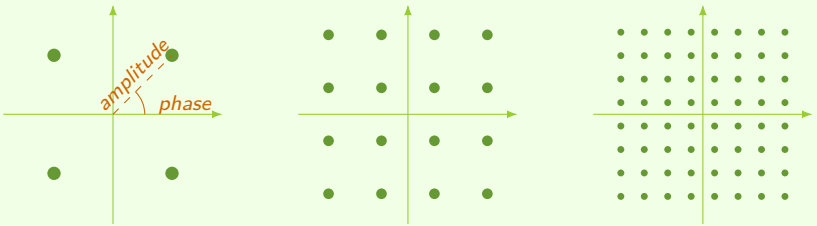
- A low-frequency baseband signal — a voice, a square wave of bits — is carried around a higher carrier frequency. The information is in the envelope.
- Why: an antenna must be a fraction of a wavelength; the regulator assigns a band; different signals can share a wire in different bands.
- A voice line passes only 300–3400 Hz: anything sent over it, data included, has to live in that window.

Amplitude, frequency, phase



- ASK: two amplitudes, here carrier or nothing. FSK: two tones. PSK: the carrier shifted by 0 or 180 degrees.
- Four phases carry 2 bits per symbol: QPSK. Frequency and phase cannot vary independently, so real schemes combine amplitude and phase.

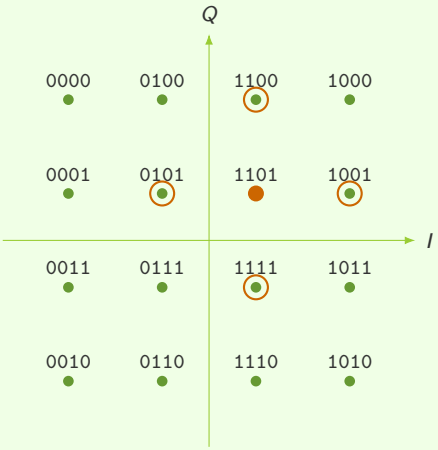
Constellation diagrams



QPSK — 2 bits/symbol QAM-16 — 4 bits/symbol QAM-64 — 6 bits/symbol

- Each dot is a legal symbol: distance from the origin is the amplitude, the angle is the phase. n dots, $\log_2 n$ bits per symbol.
- Square grids, because electronics set a value on each axis more easily than an amplitude and an angle.
- Denser grids need more S/N to keep the dots apart: this is where Shannon's budget is spent.

One wrong point, how many wrong bits?



Noise pushes the symbol 1101 onto a neighbouring point. How many bits are wrong?

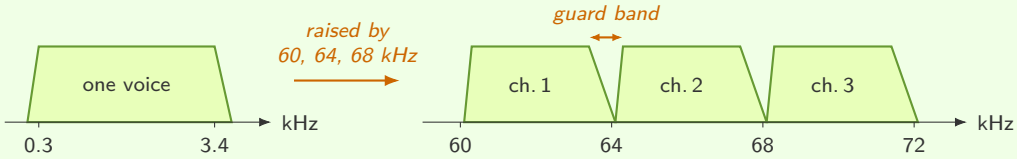
Numbered in counting order, a neighbour of 0111 could be 1000: four bits wrong. With a Gray code every neighbour differs in one bit.

- The neighbours of 1101 are 1100, 1001, 1111 and 0101: one wrong bit each.
- Noise usually moves a symbol only a little, so the common error costs one bit.

Why share one line

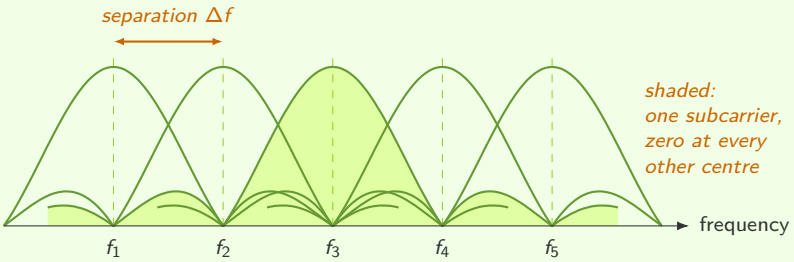
- A line costs about the same whatever its capacity: the money goes into digging the trench, not into what is laid in it.
- So one high-capacity line is shared by many signals: multiplexing. By frequency, by time, by code — and by wavelength on a fibre.
- We met the same names as ways for stations to share a broadcast channel: multiple access.
- The difference is who does the sharing. In multiplexing one device, the multiplexer, combines the streams at one end and separates them at the other.
- In multiple access the stations are independent and must coordinate over the channel itself. Same division, different problem.

Frequency division and guard bands



- Each voice channel, filtered to 300–3400 Hz, is raised by a different amount; no two now overlap, and they can be added on one line.
- 4000 Hz per channel: the excess is a guard band, because real filters have no sharp edges.
- AM radio is the same thing in the air. In the telephone core FDM gave way to TDM; it survives in radio and cable.

OFDM



- The band is cut into many subcarriers, each sending its own data with QAM, packed with no guard bands.
- Each spectrum spills into its neighbours but is zero at their centres: sampled there, subcarriers do not interfere.
- One fast stream becomes many slow ones; a subcarrier hit by noise gets fewer bits, or none. 802.11, 4G, cable, and ADSL.

Time division, fixed and statistical

- TDM: streams take turns, round-robin, each owning the whole band for a fixed slot. The ends must stay synchronised; short guard times absorb the drift.
- Statistical TDM: slots go by demand, not by schedule — which is packet switching under another name.

Voice in the telephone core

A 4 kHz voice channel is sampled 8000 times a second — Nyquist — at 8 bits per sample: 64 kb/s. A T1 line interleaves 24 of them plus one framing bit: 193 bits every 125 μs, 1.544 Mb/s.

- Sampling runs the other way from everything so far: an analog signal becomes bits, in the codec at the exchange.

Code division: what the chips cost

- Each bit time is cut into m chips. A station sends its m-chip code for a 1, and its negation for a 0.
- With orthogonal codes, the receiver multiplies by the code it wants and every other station vanishes — the algebra we did for multiple access.
- The cost: m chips per bit take m times the bandwidth. 100 stations on 1 MHz: FDM gives each 10 kHz; CDMA gives each the whole MHz, at 100 chips per bit.
- Walsh codes: 2^n orthogonal codes of length 2^n.
- They stay orthogonal only if the chips arrive aligned in time — rarely true on a cellular network, which needs other designs.

Wavelength division

- WDM: frequency division on a fibre: each signal on its own wavelength, combined into one fibre, split apart at the far end.
- The split is made by a diffraction grating — completely passive, and therefore very reliable.
- Why it pays: one channel is held to a few GHz by electro-optical conversion, in a band of 25 000 GHz. Parallel wavelengths use the rest.
- 1990: 8 channels of 2.5 Gb/s. 2019: 160 channels and more than 16 Tb/s on one pair of fibres.
- Optical amplifiers regenerate all the wavelengths at once, without turning each one back into electricity.

Why the modem stopped at 33.6 kb/s

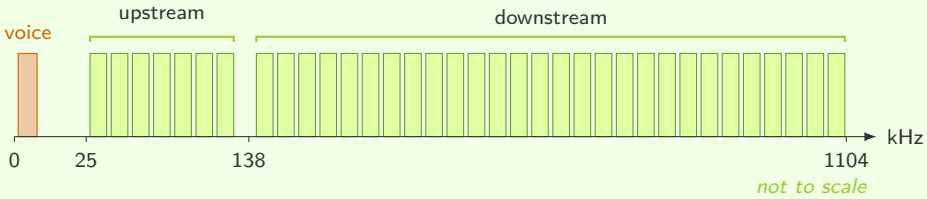
- A telephone modem sends data where the voice would go: 3100 Hz, at 2400 symbols a second.
- Denser constellations, generation after generation: 4, 6, 12, 14 bits per symbol — 9600, 14 400, 28 800, 33 600 b/s.

Why did the progression stop at 14 bits per symbol?

Shannon. Across two average local loops the voice channel's capacity is about 35 kb/s. Faster needs other physics, or other wires.

- The 56k trick: the provider connects digitally, removing one noisy analog loop. The limit becomes the exchange: 8000 samples of 7 usable bits, 56 kb/s.

ADSL: remove the filter, split the line



- The 3100 Hz were a filter at the exchange, not the wire: bypass it, and the pair offers about 1.1 MHz.
- OFDM — here called DMT — cuts it into 256 channels: voice, a few empty, upstream, and a larger downstream: the A of ADSL.
- Each channel runs QAM at about 4000 symbols a second, from 15 bits per symbol where S/N is high to none where it is not.
- Twisted pair, passband, QAM, OFDM, Shannon: all of today, on the wire into your home.

Summary

- Bits become signals — voltage, light, radio — over guided or unguided media, each with its own bandwidth, reach and weakness.
- A channel passes a band of frequencies. Nyquist bounds how fast it can signal, Shannon how many bits it can carry: $B \log_2(1 + S/N)$.
- Baseband: line codes trade bandwidth for clock recovery and balance: Manchester, NRZI, 4B/5B, scrambling, 8B/10B.
- Passband: amplitude, frequency and phase of a carrier; constellations as dense as the noise allows, Gray-coded.
- One line, many signals: by frequency, time, code and wavelength — and OFDM, which packs subcarriers with no guard bands.
- The stack is complete, from the application to the wire. Next: what happens when someone on it is hostile.

Your own link, as the adapter sees it

```
$ ethtool enp3s0 | grep -E 'Speed|Duplex|Port'
    Speed: 1000Mb/s
    Duplex: Full
    Port: Twisted Pair
$ iw dev wlp2s0 link | grep -E 'freq|signal|tx bitrate'
    freq: 5180
    signal: -52 dBm
    tx bitrate: 866.7 MBit/s VHT-MCS 9 80MHz short GI VHT-NSS 2
```

- Wired: a gigabit on four twisted pairs, full duplex, negotiated by the two adapters.
- 80MHz: the analog bandwidth, the B of Shannon. $-52\,\mathrm{dBm}$: the received power, about 6 nanowatts.
- MCS 9: 256-QAM on OFDM subcarriers, two spatial streams.
- Walk away from the access point and watch the MCS fall: the adapter picks the densest constellation the S/N allows.